Study of Higgs Boson Mass Reconstruction in the $tHq(H \to \tau\tau)$ Production with Two Light Leptons and a Hadronically Decaying Tau Lepton – Summer Student Report

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Abstract

This project aims to reconstruct the mass of the Higgs boson using simulated events of the type $tHq(H \to \tau\tau)$. A dedicated deep neural network is implemented to predict the Higgs boson mass using information from assigned light leptons and other event features. This approach focuses to enhance the signal efficiency and to suppress background reactions.

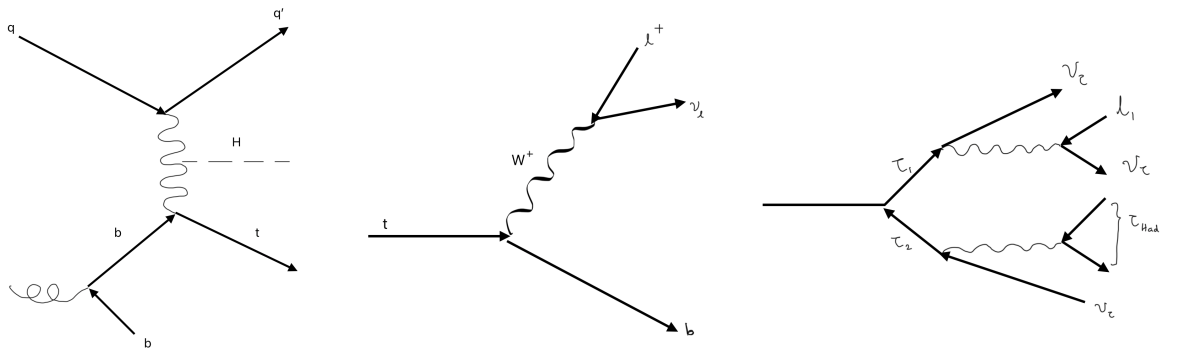
1 Introduction

1.1 Higgs boson and Standard Model

The Standard Model (SM) of particle physics represents our best understanding of physics phenomena at the most fundamental scales. It provides a unified picture for all known elementary particles, the building blocks of matter and the way they interact via three of the four fundamental forces. Over the decades, the SM has been tested extensively by a broad variety of experiments. It is able to successfully explain almost all experimental results over a wide energy range with a great precision. Its remarkable predictive power was once again exceptionally demonstrated by the discovery of the Higgs boson in 2012 at CERN's Large Hadron Collider (LHC). At the LHC, the Higgs boson is produced in several production channels. Further details on this project are given in Ref. [1].

1.2 Higgs Decays and tHq Production

The top quark is the heaviest elementary particle in the SM. After the discovery of a Higgs boson, the properties of the Higgs boson must be examined to compare them with SM predictions. One such property is the Yukawa coupling (Y_t^{SM}) between the top quark and the Higgs boson. This coupling can be measured directly in processes where both the Higgs boson and real top quarks are present. There are two processes that allow a direct measurement of Y_t^{SM}. The first is the production of the Higgs boson with a top anti-top pair ($t\bar{t}H$), while the second is the production of the Higgs boson with a single top quark (tHq) [2]. Decays of the Higgs boson can occur in many ways. In this study, the H $\rightarrow \tau\tau$ decay is addressed. The tau decays leptonically and hadronically. Hence, the channel can be narrowed to two light leptons and one hadronically decaying tau ($2\ell + 1\tau_h$).Figures 1a and 1b show the production of tH and the decays of the top and Higgs bosons [2].



(a) tH production

(b) t and H decays

Figure 1: Feynman diagrams showing (a) tH production and (b) t and H decays

1.3 The ATLAS Detector

The ATLAS detector is a multipurpose particle detector built around the LHC. The primary objective of this detector is to address the inquiries posed in the SM and Beyond the Standard Model (BSM). It searches for new physics beyond the SM. The ATLAS detector (Figure 2) measures properties of particles, such as their direction, momentum, charge, energy, and particle type [3]. And it also investigates the properties of the Higgs boson. As the largest volume particle detector at the LHC, it comprises numerous sub-detectors that identify and measure particles produced in proton-proton collisions. The ATLAS experiment has made a significant contribution to the advancement of modern particle physics, particularly regarding the Higgs boson [3]. Previously, initial studies on the Higgs boson mass reconstruction were carried out for the ttH and tHq processes [4]. A challenge when reconstructing the mass is the missing energy due to the undetectable particles. For example, neutrinos cannot be detected by the ATLAS detector [3].

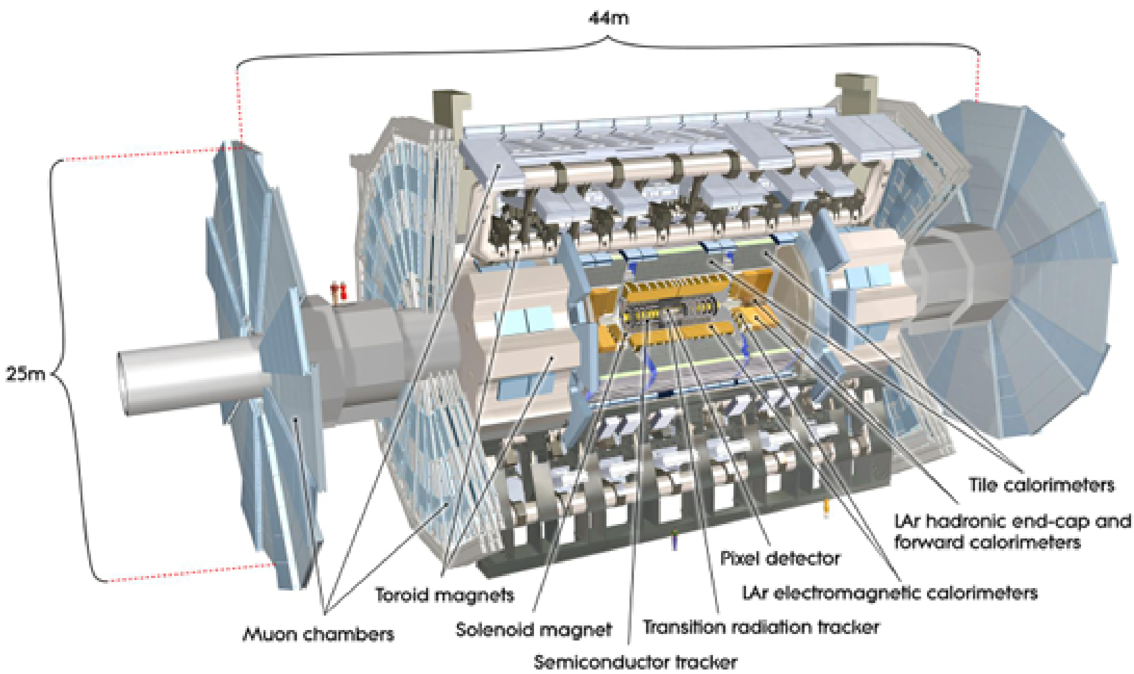


Figure 2: Cut-away view of the Atlas Detector

1.4 Artificial Neural Network

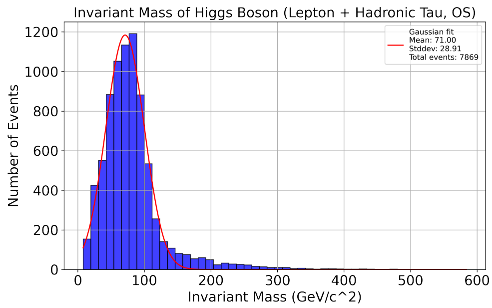
With the development of Machine Learning, a novel method can be applied to reconstruct the mass of the particle and to separate the background and the signal events with a greater precision. The construction of a Neural Network entails several distinct processes, including the formation of its layers. These layers comprise the input layers, the output layers, and a layer referred to as the hidden layer. A Regression Neural Network (RNN) can be implemented to predict a single value or multiple values, such as the mass of particles.

2 Analysis

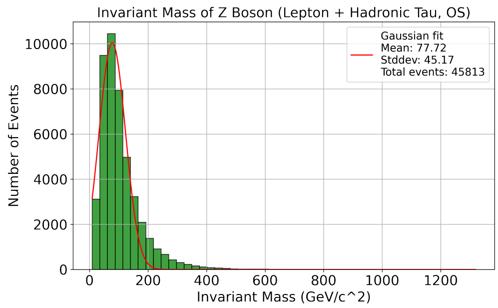
In the $2\ell + 1\tau_h$ channel, two cases are considered, Opposite Sign (OS) leptons and the Same Sign (SS) leptons.

2.1 The Invariant Mass

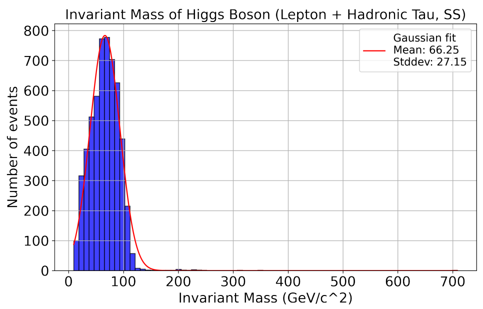
Figures 3a and 3b, and Figures 3c and 3d show the invariant mass distributions for the Higgs boson and Z boson, respectively.



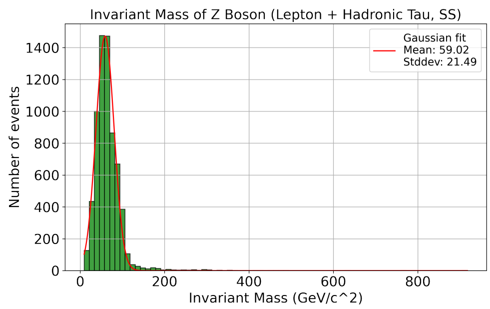
(a) tH invariant mass for Opposite Sign leptons



(b) tZ invariant mass for Opposite Sign leptons



(c) tH invariant mass for Same Sign leptons

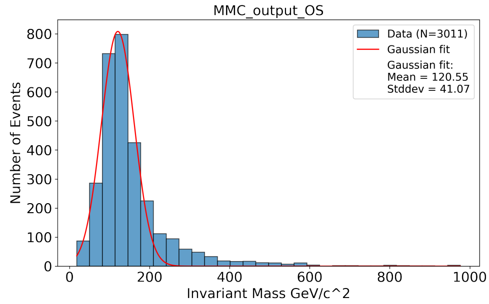


(d) tZ invariant mass for Same Sign leptons

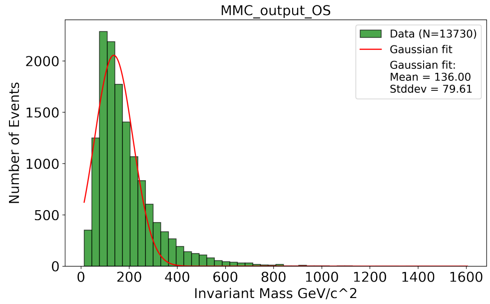
Figure 3: Invariant mass distribution for tH and tZ in different lepton configurations.

2.2 MMC Application

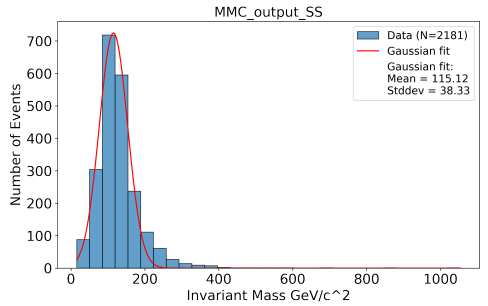
The Missing Mass Calculator (MMC) method was used for comparison to show the reconstructed mass of the Higgs boson and the Z boson. It outperforms other common techniques, such as the invariant mass method [5]. Figures 4a and 4b, and Figures 4c and 4d show the reconstructed mass for the Higgs boson and Z boson, respectively.



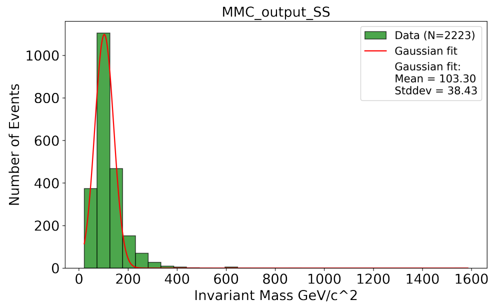
(a) Opposite sign tH after $m > 0$ cut



(b) Opposite sign tZ after $m > 0$ cut



(c) Same sign tH after $m > 0$ cut

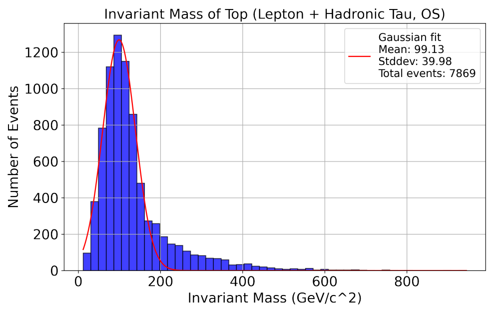


(d) Same sign tZ after $m > 0$ cut.

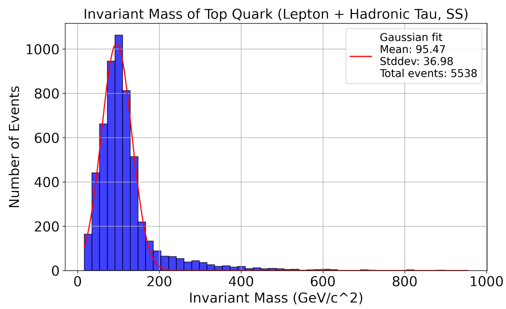
Figure 4: MMC reconstructed mass for the Higgs boson and Z boson with Opposite Sign leptons and Same Sign leptons

2.3 Top Mass Reconstruction

Figures 5a and 5b show the reconstructed mass of the top quark for opposite sign and same sign leptons, respectively.



(a) Reconstructed mass for the top quark (Opposite Sign leptons)

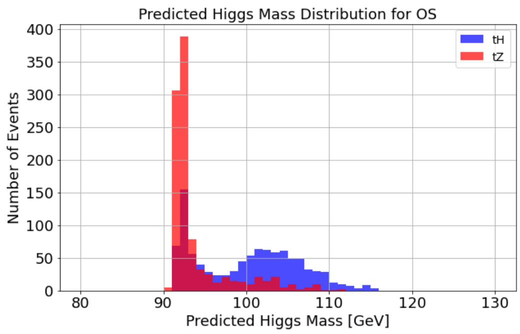


(b) Reconstructed mass for the top quark (Same Sign leptons)

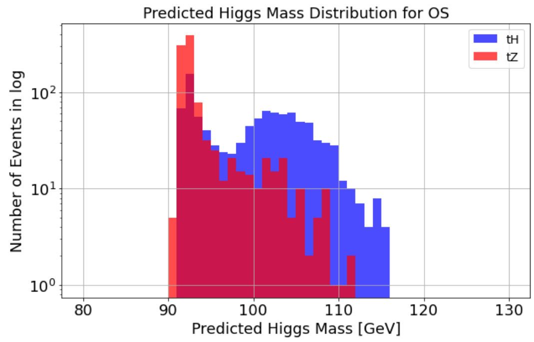
Figure 5: Reconstructed mass distribution for the top quark in different lepton configurations.

2.4 Neural Network Results

An RNN was implemented using the simulated data for tH and tZ. Then the NN was tested for the OS leptons and SS leptons separately. Figures 6 and 7 show the results of the Neural Network for Opposite Sign and Same Sign leptons, respectively.

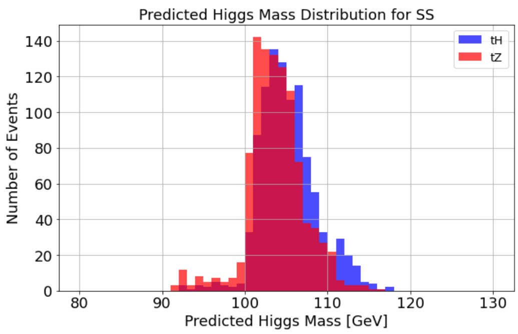


(a) Opposite Sign leptons (linear)

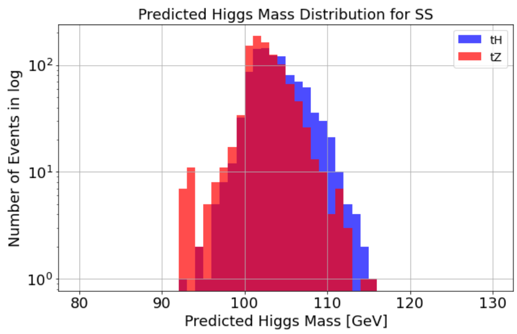


(b) Opposite Sign leptons (logarithmic)

Figure 6: Neural Network results for tH and tZ (Opposite Sign leptons).



(a) Same Sign leptons (linear)

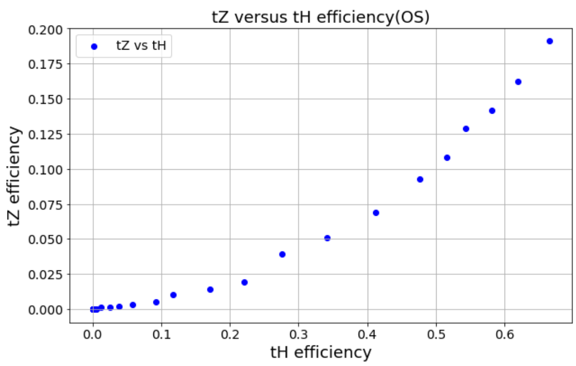


(b) Same Sign leptons (logarithmic)

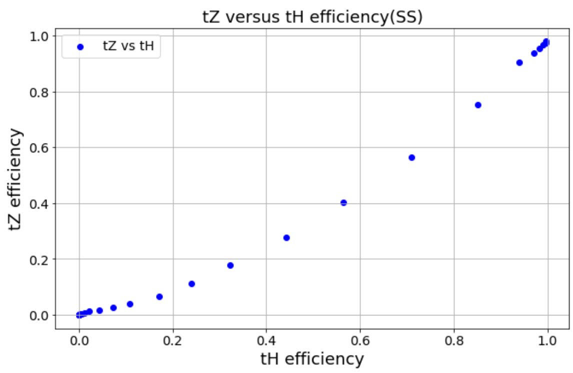
Figure 7: Neural Network results for tH and tZ (Same Sign leptons)

2.5 Performance

To quantify the performance of the NN, the tH and tZ efficiency were used as a metric. The efficiencies of tH and tZ are correlated. Figure 8 shows the efficiency curve for Opposite Sign (OS) and Same Sign (SS) leptons.



(a) Efficiency curve (Opposite Sign)



(b) Efficiency curve (Same Sign)

Figure 8: Efficiency curves for Opposite Sign (OS) and Same Sign (SS) leptons

3 Conclusions

In this study, a Neural Network model was applied to predict the mass of the Higgs boson in $tHq(H \to \tau\tau)$ production. The results highlight how Neural Networks can effectively handle highly complex and nonlinear problems. While the model performed well and achieved notable accuracy for the opposite sign leptons, it performs much weaker for the same sign leptons. Enhancing the model could involve adding more relevant features, fine-tuning the hyperparameters, using larger and more detailed datasets, and using truth information for tH and tZ events in the training of the Neural Network.

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References

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